Data Flow Diagram

Date	06-07-2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

A Data Flow Diagram (DFD) to show how data moves through the OptiCrop system, between the external entity, the prediction process, and the model data store. The symbol legend below is used as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Farmer).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Input, Predict Crop).
Rectangle (open, no number)	Data Store	A place where data is held for later use (e.g., Trained ML Model).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Level 0 Data Flow Diagram

Project: Smart Agricultural Production Optimization Engine

Main Process:

• 1.0 Input Collection — collects soil and climate parameters submitted by the farmer through the web form.

• 2.0 Validate & Scale Input — validates the entered values and applies the fitted StandardScaler.

• 3.0 Predict Crop — runs the trained Logistic Regression model on the scaled input to predict the most suitable crop.

• 4.0 Display Result — renders the predicted crop recommendation on the result page.

External Entity:

• Farmer — provides soil and climate parameters and receives the crop recommendation.

Data Store:

• D1: Trained ML Model — stores the trained Logistic Regression model (crop_model.pkl) and the fitted StandardScaler (scaler.pkl), loaded once at application startup.

Data Flows:

• Farmer → 1.0 Input Collection : Soil & Climate Parameters (N, P, K, Temperature, Humidity, pH, Rainfall)

• 1.0 Input Collection → 2.0 Validate & Scale Input : Raw Input Values

• 2.0 Validate & Scale Input → 3.0 Predict Crop : Scaled Feature Vector

• D1 → 3.0 Predict Crop : Load Trained Model & Scaler

• 3.0 Predict Crop → 4.0 Display Result : Predicted Crop Label

• 4.0 Display Result → Farmer : Crop Recommendation Displayed

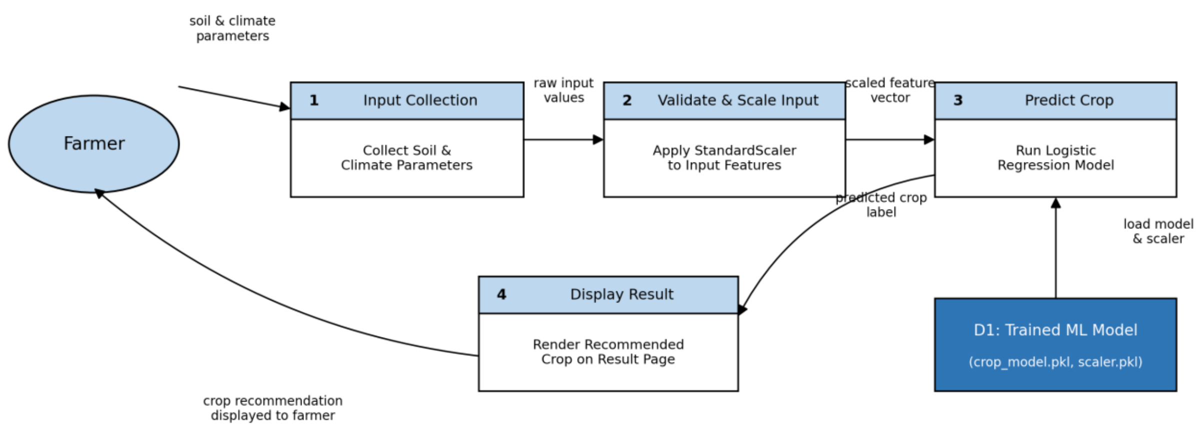

Figure 1: Level 0 DFD for Smart Agricultural Production Optimization Engine